CHAPTER 1

Syllabus Map & Strategy

GATE 2027 CN syllabus · what is in, what is out · priority order · how these notes are organised

QUICK TRICK

The whole GATE 2027 Computer Networks syllabus is ONE line. Learn it by heart: Layering · Switching + performance metrics · Error detection · MAC · Ethernet · Distance vector & link state · IPv4 fragmentation, CIDR, NAT · TCP flow & congestion control, socket API · DNS & HTTP.
About **8 marks** per paper on average (range ≈ 6–11). Most of it is **calculation**, so practise numbers, not definitions.

1.1 The official syllabus (GATE 2027, IIT Madras)

● **LEVEL 1 · BEGINNER**

Section 10 of the CS paper, word for word from the official syllabus PDF (gate2027.iitm.ac.in):

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Principles of Layering;
Basics of switching (circuit, packet and virtual circuit) and performance metrics;
Data link layer: error detection, Medium Access Control, Ethernet;
Distance vector and link state routing;
IPv4 - Fragmentation, CIDR Notation, Network Address Translation;
TCP - flow control and congestion control, socket API;
DNS and HTTP.
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Always confirm the final wording on the official site before the exam; syllabi can be revised. Some older blogs still show the 2026 list, so trust the official PDF.

1.2 What changed from 2026

● **LEVEL 2 · EXAM**

2026 wording	2027 wording	Meaning for you
OSI and TCP/IP stacks	Principles of Layering	Encapsulation, service/interface/protocol, addresses per layer. Layer-name trivia is low priority.
Packet, circuit, VC switching	+ performance metrics	Delays, RTT, throughput, bandwidth-delay product are now named explicitly.
Framing, error detection, MAC, Ethernet bridging	Error detection, MAC, Ethernet	Framing and bridge learning/spanning tree are not named.
Shortest path, flooding, DV, LS	Distance vector and link state	Stand-alone flooding and shortest-path are not named (Dijkstra is still used inside link state).
ARP, DHCP, ICMP	removed	Low priority. Short sheet in Chapter 16.
UDP, TCP, sockets	TCP flow/congestion control, socket API	UDP not named, but it appears as contrast in socket questions.
DNS, SMTP, HTTP, FTP, Email	DNS and HTTP	SMTP, FTP, email are not named.

Removed topics can still hide *inside* a retained topic (for example UDP under DNS, or the ICMP TTL field in an IPv4 question). So keep one revision page for each; that is Chapter 16.

1.3 Where the marks come from

● **LEVEL 2 · EXAM**

Priority tiers below are an **analyst estimate** from past papers (not an official statement). Use them to order your time.

Tier	Topics	Notes chapter
Tier 1 (very high)	TCP congestion control · CIDR and subnetting · sliding-window ARQ arithmetic · delay arithmetic and performance metrics	13 · 9 · 5 · 3
Tier 2 (high)	IPv4 fragmentation · distance vector · CSMA/CD calculations · TCP connection management · CRC · DNS resolution · switching modes · longest-prefix match · TCP flow control	10 · 8 · 6 · 12 · 4 · 15 · 3 · 10 · 13
Tier 3	NAT tables · HTTP RTT counting · Ethernet frame sizes · socket call order · layering principles	11 · 15 · 7 · 14 · 2

GATE POINT

Recent papers: most CN **2-mark** questions are **NAT** (numerical answer type); statement questions often come as **MSQ** (no partial credit). Neither has negative marking, so always attempt.

1.4 Chapter map of these notes

● LEVEL 1 · BEGINNER

Ch	Title	Syllabus clause covered
2	Principles of Layering	Principles of layering
3	Switching & Performance Metrics	Switching + performance metrics
4	Error Detection	Data link: error detection
5	Sliding Window Protocols	Data link flow/error control (used in every ARQ numerical)
6	Medium Access Control	Data link: MAC
7	Ethernet	Data link: Ethernet
8	Routing: Distance Vector & Link State	DV and LS routing
9	IPv4 Addressing, CIDR & Subnetting	IPv4, CIDR notation
10	IPv4 Header, Fragmentation & Forwarding	IPv4 fragmentation
11	Network Address Translation	NAT
12	TCP Segment & Connection Management	TCP (basis for flow/congestion control)
13	TCP Flow & Congestion Control	TCP flow control and congestion control
14	Socket API & UDP	Socket API
15	DNS & HTTP	DNS and HTTP
16	Out-of-Syllabus Quick Sheets	ARP, DHCP, ICMP, SMTP/FTP, framing, bridging, OSI
17	Master Formula Sheet & Mixed Practice	Revision

1.5 Study order and strategy

● LEVEL 2 · EXAM

1. Delay arithmetic (Ch 3) → 2. Sliding window (Ch 5) → 3. IPv4: CIDR, fragmentation, NAT (Ch 9–11) → 4. TCP (Ch 12–14) → 5. CRC, CSMA/CD, Ethernet (Ch 4, 6, 7) → 6. Routing (Ch 8) → 7. DNS and HTTP (Ch 15) → 8. Chapters 2 and 16 as light reading.

QUICK TRICK

The bandwidth-delay product (Ch 3) sits under sliding windows, TCP windows and sequence-number sizing. Master it first and three chapters get easier.

COMMON MISTAKE

• Treating CN as pure theory. Almost every 2-mark question is a calculation with units (bits vs bytes, ms vs s, Mbps = 10^6 bps).
• Dropping a removed topic completely. Keep one page of notes on each.
• Reading 1 KB as 1000 or 1024 without checking the question. Recent papers state the convention.

1.6 Books and sources

Kurose & Ross (top-down, best for TCP, DNS, HTTP) · Forouzan (Data Communications and Networking: subnetting and numericals) · Tanenbaum (CRC, CSMA/CD, sliding window) · Stallings (ARQ efficiency formulas). Practise with previous-year GATE questions, then re-tag them as in-syllabus or low-priority using 1.2.

Next: Chapter 2 — Principles of Layering (encapsulation, service vs protocol, which address works where).

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